

Q-Sched: Pushing the Boundaries of Few-Step Diffusion Models with Quantization-Aware Scheduling

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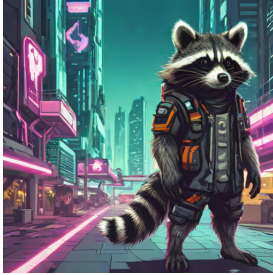
FP16 DiT Memory: 4.76 GB ELO: 1008.3  FLUX.1 [schnell] Prompt: tropical gelato glistened with gold	SVDQuant W4A4 + LoRA DiT Memory: 1.24 GB ELO: 1001.1 	Q-Sched W4A8 DiT Memory: 1.19 GB ELO: 1002.3 
FP16 U-Net Memory: 1.03 GB ELO: 999.3  SDXL-Turbo (4-Step) Prompt: cute raccoon in cyberpunk attire, standing in front of a futuristic landscape, hyper detailed	MixDQ W8A8 U-Net Memory: 0.257 GB ELO: 998.7 	Q-Sched W8A8 U-Net Memory: 0.257 GB ELO: 1000.3 

Figure 1. When large diffusion model backbones are compressed to 8-bit or 4-bit settings for deployment, image fidelity drops. Q-Sched applies scheduler-level tuning to steer the sampler back to FP16-like quality, with no new checkpoints finetuning, or extra FLOPs.

Abstract

Text-to-image diffusion models remain computationally intensive: generating a single image typically requires dozens of passes through large transformer backbones (e.g., SDXL uses ~ 50 evaluations of a 2.6B-parameter model). Few-step variants reduce the step count to 2–8 but still rely on large, full-precision backbones, making inference impractical on resource-constrained platforms. Existing post-training quantization (PTQ) methods are further hampered by their dependence on full-precision calibration.

We introduce Q-Sched, a scheduler-level PTQ approach that adapts the diffusion sampler while keeping the quantized weights fixed. By adjusting the few-step sampling trajectory with quantization-aware preconditioning coefficients, Q-Sched matches or surpasses full-precision quality while delivering a $4\times$ reduction in model size and preserving a single reusable checkpoint across bit-widths. To learn these coefficients, we propose a reference-free Joint Alignment–Quality (JAQ) loss, which combines text–image compatibility with an image-quality objective for fine-grained control; JAQ requires only a handful of calibration prompts and avoids any full-precision inference during calibration.

Empirically, Q-Sched yields substantial gains: a **15.5%** FID improvement over the FP16 4-step Latent Consistency Model and a **16.6%** im-

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provement over the FP16 8-step Phased Consistency Model, demonstrating that quantization and few-step distillation are complementary for high-fidelity generation. A large-scale user study with 80,000+ annotations further validates these results on both FLUX.1[schnell] and SDXL-Turbo. Our code is available at <https://github.com/enyagroup/q-sched>.

1. Introduction

Diffusion models have firmly established state-of-the-art performance across modalities, delivering unprecedented fidelity in image synthesis (Esser et al., 2024; Dai et al., 2023), video generation (Brooks et al., 2024; Polyak et al., 2024), and audio modeling (Liu et al., 2024). Beyond creative media, they have become indispensable in scientific discovery, driving breakthroughs in biomolecular structure prediction (Abramson et al., 2024; Watson et al., 2023). However, these capabilities come with significant computational overhead. Leading foundation models such as Stable Diffusion 3 (Esser et al., 2024) and Sora (Brooks et al., 2024) rely on massive Diffusion Transformer (DiT) backbones (Peebles & Xie, 2023), where inference necessitates iterative denoising through tens of billions of parameters, making real-time deployment prohibitively expensive.

Practical deployment therefore hinges on two levers: (1) reducing the number of function evaluations (few-step sampling), and (2) lowering the cost per evaluation (compression via quantization (He et al., 2024; Guo et al., 2022), pruning (Fang et al., 2024), or distillation (Huang et al., 2024)). These levers are particularly important in two widely used settings. *On-device*, memory and compute budgets are tight, latency and energy constraints are strict, and privacy/offline use cases preclude server offloading (Zhao et al., 2024b). *In data centers with MIG partitioning*, a single GPU is sliced into multiple smaller instances to increase concurrency and predictability; each slice has limited memory/throughput, making model footprint and per-step cost decisive (Zhang et al., 2023; Li et al., 2022). In both cases, few-step sampling and quantization are natural, complementary choices.

However, few-step acceleration is sensitive to the accuracy of the underlying probability-flow ordinary differential equation (ODE) or variance-preserving stochastic differential equation (SDE) that links the noise-estimation network to the final sample (Song et al., 2021). Quantization perturbs that network, inducing a mismatch that alters the ODE/SDE trajectory and amplifies artifacts, an effect that becomes more pronounced as the number of steps shrinks. Simply reusing full-precision schedulers on quantized backbones will inevitably induce quality degradation.

To bridge this gap, we introduce **Q-Sched**, a quantization-

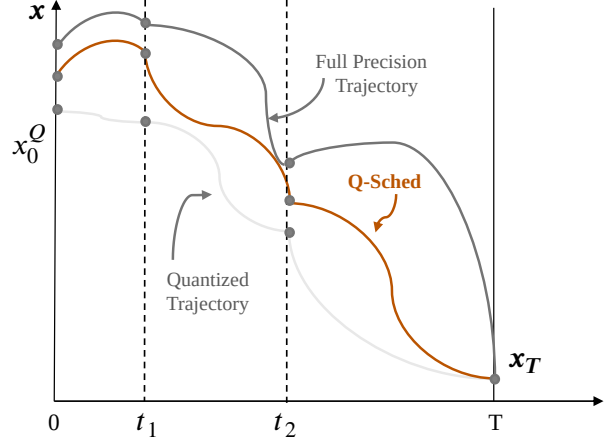


Figure 2. Quantization shifts the diffusion sampling trajectory, reducing image quality and departing from the full precision model’s sampling trajectory. Q-Sched corrects this drift by adapting the sampling scheduler.

aware noise scheduler that adapts the few-step trajectory to the compressed model *while keeping quantized weights fixed*. Instead of correcting the model weights, Q-Sched inserts lightweight coefficients (c^x, c^e) into the scheduler (Figures 2 and 3), correcting quantization-induced drift while keeping a single U-Net/DiT checkpoint reusable across multiple precision settings. This design directly targets the constraints above: it preserves the latency benefits of few-step sampling, fits within on-device and MIG memory budgets, and avoids checkpoint sprawl in production.

Our contributions are summarized as follows:

1. In this work, we introduce **Q-Sched**, a quantization-aware scheduler that integrates seamlessly with few-step diffusion models. It achieves up to a **15.5% FID improvement** over a 4-step latent consistency model (LCM) (Luo et al., 2023) baseline and, as shown in Figure 1, can match or surpass full-precision arena scores *while simultaneously reducing model size* on SDXL-Turbo (4-Step) (Sauer et al., 2024) and FLUX.1[schnell] (Black Forest Labs, 2024).
2. Q-Sched’s novel **preconditioning coefficients** enable quantized models to deliberately deviate from potentially overfit few-step baselines (Figure 2), alleviating oversmoothing and texture artifacts from distillation and quantization while improving the balance between fidelity and artifact severity.
3. To optimize these coefficients, we propose the **Joint Alignment-Quality (JAQ) loss** which balances perceptual fidelity with text-image alignment. Being reference-free, JAQ also enables precise control over visual properties (*e.g.*, texture, detail, saturation) without requiring access to a full-precision model.

4. We establish a **theoretical existence guarantee** (Theorem 3.1), proving that there exists Q-Sched coefficients which reduce expected sampling error relative to the original quantized scheduler. This provides a principled explanation for Q-Sched’s systematic improvements.
5. Finally, a large-scale **human preference study** with over 80,000 annotations demonstrates that Q-Sched outperforms MixDQ (Zhao et al., 2024a) on SDXL-Turbo and SVDQuant (Li et al., 2025) on FLUX.1[schnell] in terms of perceived image quality.

As illustrated in Figure 1, Q-Sched attains the highest ELO rating in pairwise image-quality comparisons among evaluated methods. Furthermore, Figure 4 shows that Q-Sched is Pareto-optimal with respect to both ELO and model size, underscoring its ability to balance perceptual quality and efficiency more effectively than competing approaches.

2. Background and Related Work

Diffusion models generate samples by denoising corrupted data across a trajectory of timesteps $t \in [0, T]$, where T is typically large (≥ 25). Each step applies a denoising network $\mathcal{E}_\theta$, conditioned on both t and its noisy input x_t . While iterative denoising yields high-fidelity samples, invoking a large U-Net or DiT backbone across many timesteps makes inference prohibitively slow in deployment. Few-step models are an important distillation strategy to address slow diffusion model inference.

One such few-step model, **SDXL-Turbo**, leverages Adversarial Diffusion Distillation (ADD), combining score distillation with an adversarial loss, to reduce sampling to just 1–4 steps, enabling real-time generation on commodity GPUs (Sauer et al., 2024). More recently, **FLUX.1[schnell]** introduces a 12B-parameter rectified-flow transformer with open weights, optimized for 1-4 step inference, making it attractive for latency-constrained serving (Black Forest Labs, 2024). Most recently, **FLUX.1[kontext]** extends the family beyond text-to-image toward *in-context* generation and editing, accepting text and images jointly and unifying both tasks in a flow-matching framework (Labs et al., 2025). These advances exemplify the field’s shift toward *deployment-ready* diffusion systems that meet strict latency and memory budgets.

Few-step diffusion and distillation. Few-step methods compress the teacher’s long trajectory into a handful of evaluations, preserving most of the fidelity at a fraction of the cost. Distillation is the primary approach: early demonstrations distilled long-run teachers into 1–8 step students, such as Instaflow (Liu et al., 2023), rectified-flow straightening (Liu et al., 2022), and adversarially guided

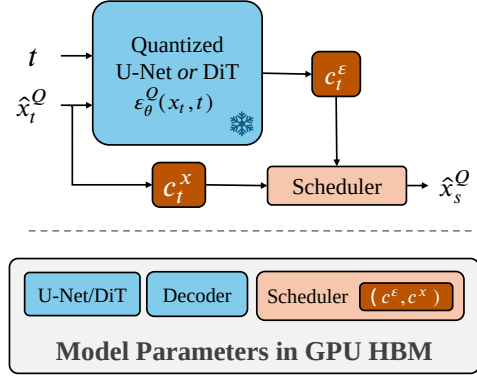


Figure 3. At timestep $t \rightarrow s$, Q-Sched applies lightweight coefficients (c_t^x, c_t^E) within the scheduler, enabling deployment of quantized models from a single U-Net or DiT checkpoint. Unlike prior approaches that alter the model backbone through retraining or post-training adjustments, Q-Sched leaves weights fixed, allowing reuse of one pretrained checkpoint across multiple precision settings. This simplifies model management and reduces storage overhead while preserving image fidelity.

ADD (Sauer et al., 2024). Consistency Models (CMs) (Song et al., 2023) frame generation as a self-consistency mapping from any noisy state to the clean sample, yielding efficient few-step samplers. Variants include Latent Consistency Models (LCMs) (Luo et al., 2023) with Stable Diffusion (Rombach et al., 2022) backbones, Trajectory Consistency Distillation (TCD) (Zheng et al., 2024) with trajectory-aware schedules, and Phased Consistency Models (PCMs) (Wang et al., 2024) with improved guidance and stability. Across these designs, the *scheduler* plays a critical role in determining quality in the few-step regime. The update rule for few-step diffusion models using quantized backbone $\mathcal{E}_\theta^Q$ is:

$$x_s = \Phi(t, x_t, \mathcal{E}_\theta^Q), \quad (1)$$

where x_s denotes the intermediate sample at timestep $s \in [0, t]$ and $\Phi(\cdot)$ is a few-step scheduler. In Section 3, we illustrate our approach using the TCD scheduler (Zheng et al., 2024) as a running example. However, Q-Sched is fully general and can be applied on top of any few-step scheduler that fits the abstraction in Equation (1).

Quantization for diffusion models. Post-training quantization (PTQ) has largely targeted $\mathcal{E}_\theta$ and its activations across timesteps. Timestep-aware calibration approaches (PTQ4DM (Shang et al., 2022), ADP-DM (Wang et al., 2023a), Q-Diffusion (Li et al., 2023)), dynamic schemes such as TDQ (So et al., 2024), and error-compensation methods (Q-DM (Li et al., 2024c)) all operate by modifying weights or activations and require full-precision calibration. MixDQ (Zhao et al., 2024a) extends to few-step models with a mixed-precision allocation strategy guided by beginning-

of-sentence(BOS)-aware quantization and layer sensitivity analysis. SVDQuant (Li et al., 2025) targets 4-bit weights and activations by absorbing outliers into a high-precision low-rank branch via SVD.

We posit that in the few-step regime, quantization bias additionally manifests as a *scheduler mismatch*, where a fixed full-precision schedule can systematically over- or under-correct and amplify artifacts. A related approach that avoids modifying network weights is PTQD (He et al., 2024), which models quantization error as an affine perturbation of the full-precision denoiser, $\mathcal{E}_\theta^Q(x_t, t) = (1 + \gamma)\mathcal{E}_\theta + \delta$, and compensates via variance scaling and a bias term in the sampler update. In PTQD, γ is estimated by standard-deviation matching, while δ is treated as uncorrelated Gaussian noise. We adapt this bias-correction principle, originally developed for undistilled diffusion models, into TCD (see Section I) and extend it to other few-step samplers as a baseline.

Crucially, Q-Sched differs from bias- and variance-scaling methods such as PTQD (He et al., 2024) by learning its correction coefficients end-to-end using final image quality, rather than relying on Gaussian assumptions or intermediate denoising states. By introducing a second coefficient on x_t and decoupling accumulated state error from current-step noise error, Q-Sched can independently correct both sources of distortion, an important property for few-step distilled models, where intermediate distributions deviate from Gaussianity. Overall, Q-Sched removes dependence on full-precision activations and directly optimizes the final output distribution.

3. Q-Sched Quantization-Aware Scheduling

Q-Sched reframes quantized few-step generation as a *scheduler adaptation* problem. It learns quantization-aware pre-conditioning coefficients that correct trajectory drift with negligible overhead, leaving the backbone frozen and requiring only lightweight calibration. Unlike prior PTQ approaches that modify weights or activations, Q-Sched adapts the scheduler itself, complementing existing PTQ and distillation techniques while retaining the latency advantages of few-step sampling.

To prepare the TCD scheduler for optimization with Q-Sched, consider sampling with a quantized network. TCD’s Strategic Stochastic Sampling (SSS) (Zheng et al., 2024) using a quantized network $\mathcal{E}_\theta^Q(x_t, t)$ is given by:

$$\mathbf{x}_s = \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{\mathbf{x}_t - \sigma_t \mathcal{E}_\theta^Q(x_t, t)}{\alpha_t} + \sigma_{s'} \mathcal{E}_\theta^Q(x_t, t) \right) + \eta \mathbf{z} \quad (2)$$

where the noise schedule is given by σ, α , and the sampler injects stochastic noise sampled from the distribution $\mathbf{z} \sim \mathcal{N}(0, I)$. The sampler relies on an intermediary timestep

$s' \in [s, t]$ where noise is added. The degree of randomness is defined by the stochastic control parameter η :

$$\eta = \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \quad (3)$$

which can be adjusted at sampling time to vary image randomness. The TCD sampler in Equation (2), used in Phased Consistency Models, is a state-of-the-art few-step diffusion method that depends on two inputs from the previous step, x_t and $\mathcal{E}_\theta^Q(x_t, t)$, which are central to applying Q-Sched.

A Learnable Schedule Pre-Conditioner To adapt the noise schedule of few-step diffusion models, Q-Sched applies two learnable scalar preconditioning coefficients, c_t^x and c_t^ϵ , applied respectively to x_t and $\mathcal{E}_\theta^Q(x_t, t)$ at time t . As illustrated in Figure 3, Q-Sched operates independently of the model backbone (U-Net or transformer), making it broadly compatible with any few-step scheduler resembling TCD as shown in Equation (4).

Under Q-Sched, the TCD sampling update becomes:

$$\begin{aligned} \mathbf{x}_s = & \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{c_t^x \mathbf{x}_t - \sigma_t c_t^\epsilon \mathcal{E}_\theta^Q(x_t, t)}{\alpha_t} + \sigma_{s'} c_t^\epsilon \mathcal{E}_\theta^Q(x_t, t) \right) \\ & + \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \mathbf{z}. \end{aligned} \quad (4)$$

Because the update rule is affine in the preconditioning coefficients, $(\mathbf{c}^x, \mathbf{c}^\epsilon) := (c_t^x, c_t^\epsilon)_{t=0}^T$, they can be fused into the existing TCD coefficients without modifying the computational graph or adding inference cost. In the TCD formulation, the estimate at the proxy timestep s' is not obtained from a separate model evaluation but is derived directly from the prediction at timestep t . As a result, the noise term associated with s' inherits the same coefficient c_t^ϵ .

To learn $(\mathbf{c}^x, \mathbf{c}^\epsilon)$, we perform a targeted hyperparameter search as outlined in Algorithm 1. In practice, we use grid search and Nevergrad’s meta-optimizer NGOpt (Rapin & Teytaud, 2018) to optimize the Q-Sched coefficients. The search space is intentionally low dimensional, involving only four coefficients per timestep across 2–8 timesteps, which makes optimization both efficient and robust. In the grid search setting, `opt.step` deterministically traverses a predefined grid and yields strong, reliable performance across models and schedules. Notably, even small coefficient adjustments consistently produce crisper images and substantially reduce quantization artifacts, highlighting the importance of scheduler-level calibration in few-step diffusion. Additional details on the Q-Sched optimization procedure are provided in Section N.

Algorithm 1 Search for Q-Sched Coefficients

Input: search range $[c_{min}, c_{max}]$, search points n , number of diffusion steps ω
 loss function JAQ, calibration set $\mathcal{C}$, search optimizer opt

- 1: Initialize $S^* \leftarrow \infty$
- 2: $\triangleright$ initialize each parameter in uniformly distributed range (c_{min}, c_{max})
- 3: $(c_{start}^x, c_{end}^x, c_{start}^\epsilon, c_{end}^\epsilon) \leftarrow \text{opt.init}(c_{min}, c_{max})$
- 4: **for** $i \in [0, n]$ **do**
- 5: $\mathbf{c}^x \leftarrow \text{linspace}(c_{start}^x, c_{end}^x, \omega)$
- 6: $\mathbf{c}^\epsilon \leftarrow \text{linspace}(c_{start}^\epsilon, c_{end}^\epsilon, \omega)$
- 7: $S \leftarrow []$
- 8: **for** $x \in \mathcal{C}$ **do**
- 9: $S_x \leftarrow \text{JAQ}(x; \mathbf{c}^x, \mathbf{c}^\epsilon)$
- 10: $S = S \cup S_x$
- 11: **end for**
- 12: $\triangleright \bar{S}$ is the arithmetic mean of S
- 13: **if** $\bar{S} < S^*$ **then**
- 14: $S^* \leftarrow \bar{S}, \mathbf{c}_*^x \leftarrow \mathbf{c}^x, \mathbf{c}_*^\epsilon \leftarrow \mathbf{c}^\epsilon$
- 15: **end if**
- 16: $(c_{start}^x, c_{end}^x, c_{start}^\epsilon, c_{end}^\epsilon) \leftarrow \text{opt.step}(\bar{S})$
- 17: **end for**
- 18: **return** $\mathbf{c}_*^x, \mathbf{c}_*^\epsilon$

A natural question arises: **why are Q-Sched coefficients sufficient to improve quantized sampling quality?** Theorem 3.1 provides theoretical justification that scheduler-level reweighting alone can strictly reduce the reconstruction error between full precision and quantized outputs at timestep $t = 0$. Specifically, letting Δx_0 denote the reconstruction error under naïve quantization, there exist scheduler coefficients $(\mathbf{c}^x, \mathbf{c}^\epsilon) \neq 0$ such that the reweighted sampler achieves lower expected reconstruction error:

Theorem 3.1 (Strict Existence Guarantees). *There exists Q-Sched coefficients $(\mathbf{c}^x, \mathbf{c}^\epsilon) \neq 0$ such that $E[||\Delta \tilde{x}_0||] < E[||\Delta x_0||]$.*

As shown in the proof in Appendix J, Δx_0 is a linear combination of per-step denoising errors $\Delta E_\theta(t)$ with coefficients k_t, m_t . Since the error is homogeneous in these terms, rescaling via $\tilde{k}_t = c_t^x k_t$ and $\tilde{m}_t = c_t^\epsilon m_t$ strictly reduces the expected error over naïve quantization. Thus, re-weighting the sampler, without modifying network weights, guarantees a reduction in error with respect to the full precision images. Next, we will discuss our reference-free metric, JAQ, and its advantages over existing PTQ loss objectives.

JAQ: A Joint Alignment Quality Loss Function Because full-precision intermediate states become unstable targets under quantization, due to structural and semantic drift, **optimizing for downstream image quality** provides a more reliable objective than attempting to match the full-precision

trajectory. Reference-free metrics such as CLIPScore (Hessel et al., 2021) are therefore particularly well suited to this setting. Unlike comparative metrics such as FID (Heusel et al., 2017) or SSIM (Wang et al., 2004), reference-free metrics do not rely on a ground-truth reference image. Under quantization, the generated image $\hat{x}_0^Q$ follows an altered sampling trajectory (Figure 9), often yielding a different and sometimes cleaner image than its full-precision counterpart. As a result, reference-based metrics fail to capture fine-grained visual differences introduced by quantization.

To better balance prompt fidelity and visual quality, we introduce the **Joint Alignment Quality (JAQ) loss**, which combines a text-to-image compatibility metric $\text{TC}(x)$ with a pure image quality metric $\text{IQ}(x)$. Unlike optimizing CLIPScore or CLIP-IQA (Wang et al., 2023b) alone, JAQ better discriminates between highly similar images, a regime where standard metrics are less informative. We define

$$\text{JAQ}(x) = \text{TC}(x) + k \cdot \text{IQ}(x), \quad (5)$$

where k controls the tradeoff between semantic alignment and visual detail. Optimizing only for text-to-image compatibility tends to sacrifice fine detail and overlook quantization artifacts (Figures 8 and 10), while image-only objectives can introduce extra details. JAQ balances these effects.

Applying Q-Sched to full precision models? While Q-Sched could in principle be applied to full precision models, it is not designed for this setting. Full precision models are better optimized through their original training objectives, such as distillation or fine tuning, rather than post-training scheduler adjustment. Q-Sched instead targets degradation introduced by quantization, correcting the resulting drift in the sampling trajectory.

4. Experiments

Experimental Setup We apply Q-Sched across few-step diffusion models, including U-Net (Ronneberger et al., 2015) and DiT (Peebles & Xie, 2023) backbones, and across various distillation strategies: consistency-based (LCM (Luo et al., 2023), PCM (Wang et al., 2024)) and flow-matching approaches (SDXL-Turbo (Sauer et al., 2024) and FLUX.1[schnell] (Black Forest Labs, 2024)). We quantize models in both 4-bit weights, 8-bit activations (W4A8) and 8-bit weights, 8-bit activations (W8A8). Only the U-Net or DiT backbone is quantized, as it dominates model size (see Table 9). Latency is measured on an Nvidia RTX A6000 GPU with Ampere compute architecture. Using BitsandBytes (Dettmers et al., 2025), we quantize each model to 4-bit weights, 8-bit activations and average latency over 10 runs with a 3-run warmup phase.

LCM and PCM are tested at 2, 4, and 8 steps on COCO-30k (Lin et al., 2014), using FID (vs. real), CLIPScore

Table 1. Comparison of scheduling strategies for Phased Consistency Models (PCMs) and Latent Consistency Models (LCMs) built on a Stable Diffusion v1.5 backbone. We evaluate the original schedules, TCD (Zheng et al., 2024) for PCMs and multi step consistency sampling (Luo et al., 2023) for LCMs, alongside PTQD (He et al., 2024) and Q-Sched. FID and CLIPScore are computed on the COCO 30k dataset. NFEs denotes the number of function evaluations, corresponding to the number of passes through the network $\mathcal{E}_\theta^Q(x_t, t)$. Latency is reported in milliseconds on an RTX A6000 GPU.

NFEs	Precision	Schedule	Latency (ms)	PCMs		LCMs	
				FID ↓	CLIPScore ↑	FID ↓	CLIPScore ↑
2	FP16	Original	148	24.17	<u>25.489</u>	38.74	<u>25.155</u>
	W4A8	Original	136	28.70	25.343	40.93	24.886
	W4A8	PTQD	137	<u>23.33</u>	25.265	<u>37.59</u>	24.919
	W4A8	Q-Sched	136	22.24	25.543	32.50	25.152
4	FP16	Original	193	23.29	25.482	<u>31.94</u>	25.969
	W4A8	Original	172	23.08	25.557	38.41	<u>25.456</u>
	W4A8	PTQD	172	<u>19.42</u>	<u>25.639</u>	39.72	24.678
	W4A8	Q-Sched	172	17.39	25.715	26.98	25.336
8	FP16	Original	286	20.15	25.714	<u>27.34</u>	26.052
	W4A8	Original	245	18.48	<u>25.664</u>	27.55	<u>25.397</u>
	W4A8	PTQD	246	15.85	25.770	28.06	25.241
	W4A8	Q-Sched	245	<u>16.83</u>	25.698	25.82	25.214

Table 2. Comparison on a 2-step Phased Consistency model using the Stable Diffusion XL backbone. FID-SD (↓) is computed relative to images generated by Stable Diffusion XL using corresponding COCO-30k prompts.

Scheduler	Precision	FID	FID-SD	CLIPScore
TCD	FP16	18.65	10.45	26.531
TCD	W4A8	22.70	12.51	26.241
PTQD	W4A8	161.96	176.29	25.910
Q-Sched	W4A8	18.89	12.17	26.513

(prompt alignment), and FID-SD (vs. Stable Diffusion). FLUX.1 and SDXL-Turbo are evaluated on the SVDQuant (Li et al., 2025; 2024b) subset of MJHQ-30k (5,000 high-quality Midjourney prompts in 10 categories), using FID and human preference studies to capture perceptual quality.

We employ two variants of the Joint Alignment Quality (JAQ) loss: one derived from CLIP-based metrics and another from human preference scores. In the CLIP-based variant, we set $TC(x) = CLIPScore(x)$ and $IQ(x) = CLIP-IQA(x)$. For SDXL-Turbo and FLUX.1, we instead adopt a preference-based variant, with $TC(x) = AQ-MAP(x)$ and $IQ(x) = HPSV2(x)$. Here, AQ-MAP (Li et al., 2024a) provides a spatial alignment score, while HPSV2 (Wu et al., 2023) is fine-tuned on real human judgments. In both cases, we fix $k = 2$.

Results: Latent and Phased Consistency Models In Table 1, we evaluate three schedulers across two consistency model families and show that Q-Sched learns a new few-

step trajectory that mitigates artifacts and can even surpass both FP16 and W4A8 in detail. It achieves strong FID scores and outperforms PTQD in 4/6 consistency variants on Stable Diffusion v1-5, while using only a fraction of the calibration set. We compare with PTQD (He et al., 2024), the only other quantization-aware scheduler for few-step diffusion. Unlike PTQD, which relies on a 1,024-image full-precision calibration set, Q-Sched requires only 20 representative sDCI prompts (Li et al., 2025), reused across evaluations. Calibration overfitting is a common issue in PTQ methods, but we mitigate it by using a highly descriptive long-form sDCI calibration set and optimizing only a small number of coefficients. This helps the quantized model learn to handle complex scenes, which generalizes well to simpler prompts. Our results show that this strategy yields strong performance on MJHQ and COCO-30k, two distinct downstream datasets. Unlike PTQD, which requires full-precision references, Q-Sched operates with just twenty prompts and can exceed a full precision few-step model by **16.1%**, **15.5%**, and **5.6%** at 2, 4, and 8 steps, respectively. This highlights that quantization and few-step distillation act as complementary compression strategies.

Furthermore, we emphasize that Q-Sched and PTQD *do not add any additional latency overhead* to the sampler, since their coefficients can be fused to the existing schedule at inference time.

In Table 2, we evaluate a large-scale 2-step Phased Consistency Model on the Stable Diffusion XL backbone. Q-Sched incurs only a 1.2% FID drop in W4A8, showing that quantization-aware preconditioning preserves quality even under aggressive compression. By contrast, PTQD

Table 3. Comparison across image quality metrics. "MixDQ" refers to the W8A8 MixDQ (Zhao et al., 2024a) variant and "SVDQ" refers to LoRA-based W4A4 SVDQuant (Li et al., 2025).

	SDXL-Turbo (4-Step)				FLUX.1 [schnell]			
	FP16	W8A8	MixDQ	Q-Sched	FP16	W4A8	SVDQ	W4A8 Q-Sched
CLIP Score $\uparrow$	25.62	25.62	25.38	25.36	25.61	25.17	25.52	25.27
CLIP IQA $\uparrow$	0.725	0.727	0.727	0.731	0.716	0.712	0.714	0.707
HPV2 $\uparrow$	0.276	0.276	0.275	0.278	0.275	0.274	0.275	0.272
AQ-MAP $\uparrow$	0.693	0.694	0.693	0.696	0.700	0.700	0.697	0.700
Pick Score $\uparrow$	18.48	18.49	18.48	18.51	18.43	18.42	18.40	18.46
MANIQA $\uparrow$	0.508	0.513	0.502	0.511	0.528	0.500	0.514	0.506
JAQ (ours) $\uparrow$	1.663	1.665	1.659	1.669	1.676	1.675	1.669	1.673

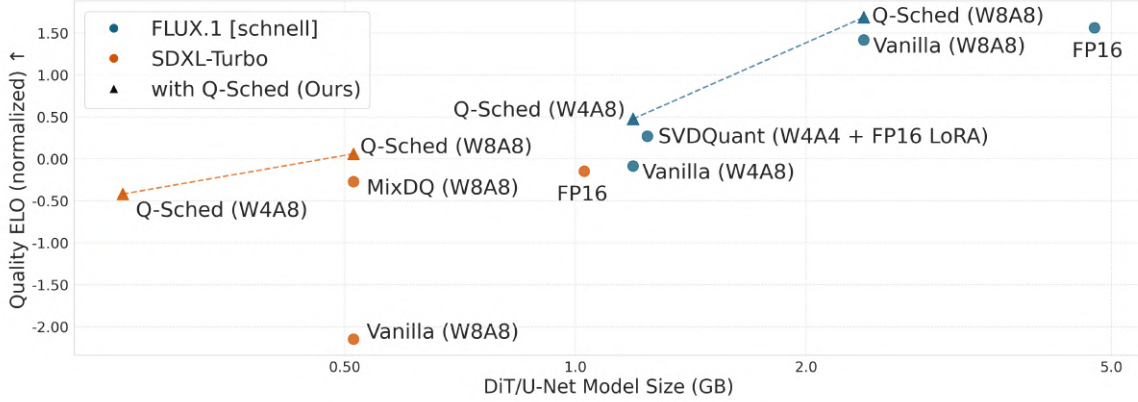


Figure 4. ELO Score vs. Model Size for various quantization methods on FLUX.1[schnell] (Black Forest Labs, 2024) and SDXL-Turbo (Sauer et al., 2024).

degrades sharply, as its Gaussian noise assumption breaks down in few-step diffusion, particularly for large models where each step approximates an ODE segment rather than a Gaussian denoising step.

Results: SDXL-Turbo and FLUX.1[schnell] We evaluate Q-Sched on FLUX.1[schnell] and SDXL-Turbo using human preference metrics, showing that JAQ effectively captures image fidelity. Beyond the metrics we’ve already mentioned, we compare with PickScore (Kirstain et al., 2023) which predicts human preferences from large-scale image-text comparisons, and MANIQA (Yang et al., 2022) which uses multi-dimensional attention to assess perceptual quality without references. As seen in Table 3, JAQ aligns closely with established metrics while uniquely balancing fine-grained details often degraded by quantization.

In Figure 4, we present user preference results for Q-Sched applied to both SDXL-Turbo and FLUX.1 [schnell], showing that Q-Sched outperforms MixDQ (Zhao et al., 2024a) and SVDQuant (Li et al., 2025), respectively, at similar model sizes (see Section C for details). We compute an ELO rating, a relative quality ranking inspired by chess scoring, by aggregating all pairwise 1v1 image comparisons across models, where a higher score reflects consistent user

preference.

In Figure 5, we compare Q-Sched across bit-widths using a user study. W4A4 proved too aggressive, but W4A5 and W4A6 produced images comparable to full precision. 1v1 comparisons with full-precision FLUX.1 (Black Forest Labs, 2024) follow the protocol in Appendix C.

Q-Sched w4a5	40.5%	59.5%	FP16
Q-Sched w4a6	43.4%	56.6%	FP16
Q-Sched w4a8	42.7%	57.3%	FP16
Q-Sched w8a8	54.3%	45.7%	FP16

User Preference (%)

Figure 5. Comparing Q-Sched across various bit-widths.

Ablation on Pre-Conditioning Coefficients We ablate the choice of pre-conditioning coefficients in the Phased Consistency Model by comparing performance when optimizing only the model-side coefficient c^e , the sample-side coefficient c^x , or both jointly. As shown in Table 4, jointly optimizing both c^e and c^x consistently yields the best results

Table 4. Ablation on pre-conditioning coefficient selection. Joint optimization of both the model and sample coefficients consistently yields the best image quality. All metrics are averaged over 1,024 images generated using a 4-step PCM with an SDXL backbone.

Metric	c^e	c^x	(c^e, c^x)
PickScore $\uparrow$	21.83	22.25	22.30
HPSV2 $\uparrow$	0.288	0.262	0.288
JAQ (ours) $\uparrow$	3.367	3.383	3.392

across all three metrics: PickScore, HPSv2, and JAQ Loss. These findings highlight the importance of treating both denoising and reconstruction terms as tunable components rather than fixing one a priori. All metrics are averaged over 1024 images generated with the SDXL backbone.

How Do We Choose k For The JAQ Loss? We optimize the Q-Sched preconditioners using the JAQ loss, which balances image quality and text-image consistency via a tradeoff hyperparameter, k . As shown in Figure 6, small k values can lead to color distortion, while larger values (e.g., $k = 5$) cause outputs to drift from the true data distribution. In such cases, the JAQ loss behaves similarly to CLIP-IQA-Q, which lacks sensitivity to concept alignment. We find that a hand-tuned value of k is sufficient for producing a high-quality noise schedule, and the final results are not highly sensitive to its exact choice. Throughout our experiments, we use $k = 2$.



Figure 6. Choice of k for the JAQ loss. k balances the contribution of $TC(x)$ vs. $IQ(x)$. Prompt: "a car and a bus on a french highway".

Choosing an Optimizer for Q-Sched We evaluate Q-Sched using grid search as well as evolutionary and meta-optimization strategies, with a subset of representative metrics shown in Table 5 and the full results reported in Table 7 in the appendix. In Algorithm 1, we intentionally keep line 16 generic to emphasize that the optimization problem is agnostic to the specific choice of optimizer. Beyond grid search, we consider Nevergrad (Rapin & Teytaud, 2018) optimizers, specifically OnePlusOne and NGOpt, as representative evolutionary and meta-optimization methods.

Overall, performance across optimizers is comparable, with

Table 5. Comparison of optimization strategies for Q-Sched on SDXL-Turbo. We use Nevergrad’s OnePlusOne and NGOpt optimizers as representative of evolutionary search and more advanced meta-optimizers.

Metric	Grid Search	OnePlusOne	NGOpt
Pick Score $\uparrow$	18.37	18.43	18.46
MANIQA $\uparrow$	0.434	0.506	0.495
JAQ (ours) $\uparrow$	1.678	1.673	1.674

differences largely within evaluation noise and no single method consistently dominating. While evolutionary strategies yield modest improvements on certain perceptual or preference-based metrics, grid search achieves strong overall performance and remains a competitive and reliable choice. These results indicate that Q-Sched is robust to the choice of optimizer, and that simple optimization strategies suffice in practice.

5. Conclusion

Few-step diffusion models dramatically reduce inference cost by distilling large generative models, such as Stable Diffusion XL, into versions requiring only 2–8 denoising steps, achieving a 5–25 $\times$ speedup. However, these models typically reduce runtime without addressing model size. Our method, Q-Sched, pushes this efficiency frontier further by introducing quantization into the few-step regime. Through noise-aware preconditioning coefficients, Q-Sched enables effective quantization with minimal performance loss. We report **8.0% and 16.1% FID improvements** over full-precision baselines for PCMs and LCMs, respectively. A user preference study also shows that **Q-Sched outperforms existing quantization methods on FLUX.1[schnell] and SDXL-Turbo in perceived image quality**. These results demonstrate that quantization and few-step distillation are complementary, enabling substantial efficiency gains without compromising generation quality.

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Impact Statement

Model compression broadens the accessibility of AI by enabling large foundation models to run on resource-constrained GPUs. The potential societal consequences of our work are similar to those of prior approaches, as both quantization and few-step diffusion serve as compression methods for text-to-image generative models. Such models can produce synthetic images that may mislead, misrepresent, or cause social harm. We conduct a user preference study on a crowdsourcing platform in which participants worldwide are shown generated content, which, like all synthetic media, carries inherent potential for misuse and harm.

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A. Q-Sched Images

In Figure 7, we provide images comparing SVDQuant with Q-Sched on a W4A4 quantized FLUX.1[schnell].

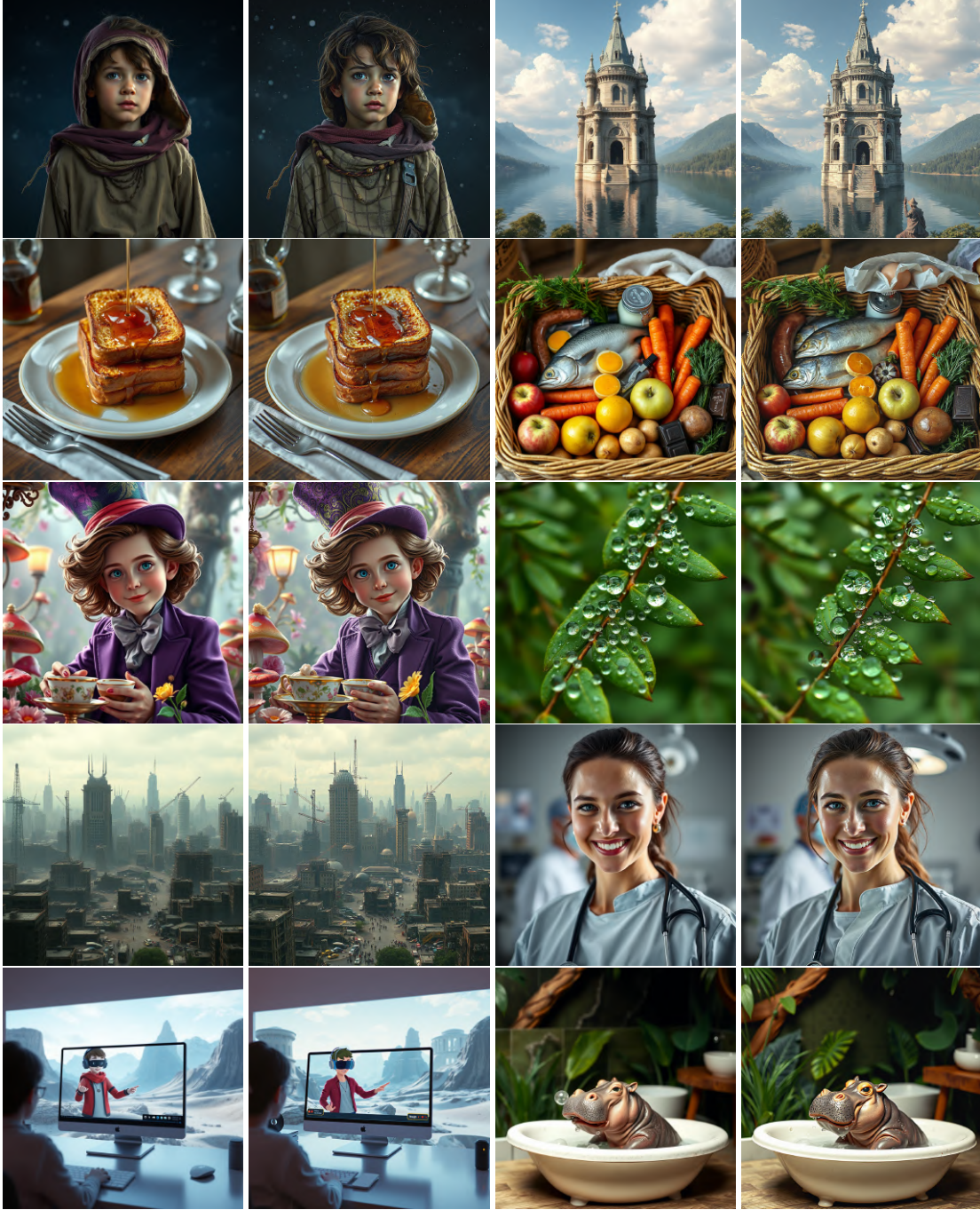


Figure 7. Side by side visual comparison of SVDQuant results and Q Sched corrected outputs for 10 MJHQ prompts. Each pair shows the SVDQuant output on the left and the corresponding Q Sched output on the right.

B. Additional Experiments on SDXL-Turbo

Table 6 reports FID on MJHQ for SDXL-Turbo under different quantization precisions and calibration strategies. Under W4A8 quantization, naive and MixDQ baselines provide limited gains, while Q-Sched substantially reduces FID from

25.75 to 21.41, outperforming the FP16 reference. This indicates that scheduler-level calibration can steer the sampler toward a different and more favorable generation trajectory, rather than merely recovering the FP16 solution. For W8A8, all methods perform similarly, suggesting that the benefits of Q-Sched are most pronounced in the low-precision regime.

More broadly, we observe that FID trends do not always align with perceptual quality or user preference metrics for modern, highly optimized diffusion models such as SDXL-Turbo, where small trajectory changes can induce distinct image distributions. In contrast, for older architectures such as consistency models, FID remains a more informative signal and we therefore report it as a primary metric in that setting. Accordingly, we present FID here as a complementary diagnostic alongside perceptual and preference-based evaluations, rather than as a sole indicator of image quality.

Table 6. FID ($\downarrow$) on MJHQ for SDXL-Turbo under different quantization precisions. The baseline FID is 25.48 for the FP16 model.

Precision	Naive	MixDQ	Q-Sched
W4A8	25.75	25.36	21.41
W8A8	25.49	25.16	26.34

Table 7. Comparison of optimization strategies for Q-Sched on SDXL-Turbo. We use Nevergrad’s OnePlusOne and NGOpt optimizers as representative of evolutionary search and meta-optimizers.

Method	CLIPScore $\uparrow$	CLIP-IQA $\uparrow$	HPV2 $\uparrow$	AQ-MAP $\uparrow$	Pick Score $\uparrow$	MANIQA $\uparrow$	JAQ (ours) $\uparrow$
Grid Search	25.65	0.709	0.277	0.701	18.37	0.434	1.678
OnePlusOne	25.17	0.713	0.274	0.700	18.43	0.506	1.673
NGOpt	25.27	0.706	0.272	0.700	18.46	0.495	1.674

C. Details on User Preference Assessment

We design our evaluation setup following the user preference study methodology from SDXL-Turbo (Sauer et al., 2024), with several improvements. For each model pair in this study, we perform 1-vs-1 comparisons based on shared prompts. Human responses, collected via Rapidata (Rapidata, 2025), come from evaluators who are presented with two images, each generated by a different model for the same prompt, and are asked: “Which image is of higher quality and more aesthetically pleasing?”

Evaluators are globally sourced and must pass a set of validation questions designed to assess annotation quality. Only those who successfully complete this qualification step are allowed to rate the models.

ELO scores are computed using the same approach as SDXL-Turbo (Sauer et al., 2024), with $K = 32$. We find that this value of K enables more noticeable ranking adjustments, especially when models have similar performance levels.

All models in our study are evaluated using 1,000 prompts sampled from the MJHQ-30k dataset. We release this subset, which we call the Q-Sched split, to enable consistent benchmarking of future quantization methods. Each prompt is evaluated by four unique annotators. Therefore, each 1-vs-1 comparison results in 4,000 total human annotations.

D. Compute Resources & Statistical Significance

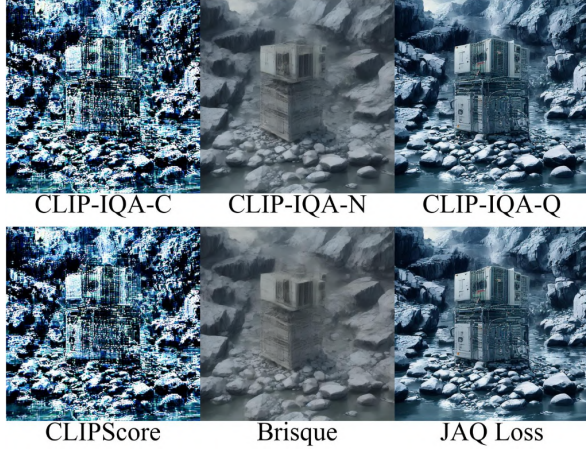
We conduct all our experiments on a high-end AI server with eight Nvidia A6000s. Each model can be run independently on one A6000 and Q-Sched takes approximately twenty minutes to run the full grid search.

Our main experiments are averaged over two-three runs but we do not report error bars at this time.

E. Ablation Studies

E.1. Comparing Loss Functions for Q-Sched

To evaluate the overall image quality for text-image generative modeling, CLIPScore (Hessel et al., 2021) is specifically designed to capture text-image compatibility and does not consider overall image quality. In Figure 8, we illustrate that Q-Sched optimized with CLIPScore produces an updated noise schedule that is over saturated and lacks image depth. In contrast, Brisque (Mittal et al., 2012) is often used as a standard reference-free image quality metric, but when used in



(a) Prompt: "An Eniac computer balanced on top of a stack of rocks over a river"



(b) Prompt: "Astronaut in a jungle, cold color palette, muted colors, detailed, 8k"

Figure 8. Optimizing Q-Sched with various reference-less image quality metrics. Our loss function, JAQ, is a linear combination of CLIPScore and CLIP-IQA-Q. We compare against three CLIP-IQA prompts: Complexity, Noisiness, and Quality denoted as -C, -N, -Q respectively.

Table 8. Adding stochasticity and its effect on W4A8 quantization for PCM using a Stable Diffusion v1-5 backbone. We report FID on COCO-30k. The stochasticity term, η , controls the amount of added Gaussian noise. $\eta = 0$ is deterministic sampling.

Method	$\eta =$					
	0	0.1	0.3	0.5	0.7	0.9
TCD	28.70	24.06	23.44	22.97	26.74	22.40
PTQD	23.33	25.59	24.95	25.69	24.53	26.71
Q-Sched	22.24	19.29	23.44	19.67	19.46	17.87

Q-Sched it creates images with smoother and less detailed features. We consider three variants of CLIP-IQA (Wang et al., 2023b) and find that CLIP-IQA using the predefined quality prompt (we denote this version by CLIP-IQA-Q) achieves a noise schedule with high-fidelity images. However, CLIP-IQA-Q has a significant weakness: it cannot properly score images with hallucinations because it does not have an understanding of the underlying image prompt or concept. Therefore, we combine the benefits of CLIPScore and CLIP-IQA-Q into the JAQ loss and find that the resulting schedule fares extremely well with respect to raw image quality as well as to concept adherence.

E.2. Adding Stochasticity

Phased Consistency Model’s implementation of the original sampler, TCD, is deterministic, meaning that there is no additive noise during sampling. The controllable noise parameter, η , allows a practitioner to adjust the additive noise during the sampling process and is defined in Equation (2). In order to compare PTQD’s correction to our method, we ablate across different levels of stochasticity and report performance for six stochasticity levels in Table 8. $\eta = 0$ refers to deterministic sampling and PTQD’s uncorrelated noise correction is not used since it adds stochastic noise by construction. Please see the appendix for more details on PTQD’s implementation in both deterministic and stochastic sampling regimes.

We find that Q-Sched outperforms PTQD for all stochasticity regimes on the 2-step phased consistency model. With learnable coefficients and no Gaussian priors, Q-Sched can outperform PTQD and the original TCD scheduler in different sampling regimes.

F. Quantization-Induced Artifacts

As shown in Figure 9, Q-Sched is able to generate images that differ sufficiently from the full precision model. We ground our quantized diffusion model with image quality metrics, rather than it’s error with respect to full preicision.

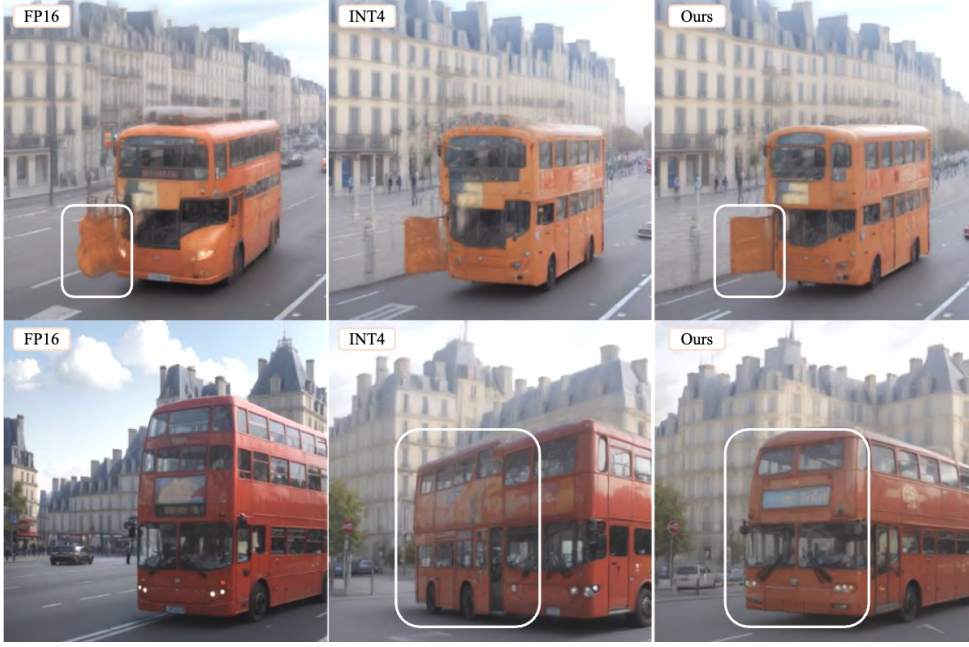


Figure 9. 4-Step (top row) and 8-Step (bottom row) LCMs. Prompt: "a car and a bus on a french highway". Q-Sched is capable of avoiding artifacts present in the FP16 or INT4 generative images. Q-Sched is close to the original schedule since it generates similar images yet our optimized schedule allows for Q-Sched to avoid some artifacts generated from the original schedule.

In our preliminary analysis using a two-step Consistency Model, we observed several characteristic ways in which quantization degrades image quality. As shown in Figure 10, quantized models tend to exhibit three prominent types of artifacts: color distortion, image degradation, and hallucinated structures. These issues are especially pronounced in low-bit settings and appear consistently across a variety of models and prompts.

G. Latency and Model Size Analysis

In Table 9, we show the full model size breakdown for the diffusion model backbone, text encoders, and the VAE decoder. During inference, either one or both text encoders are used, and we do not need the VAE encoder, since this is for training exclusively.

For our ELO vs. Model Size Pareto front in Figure 4, we consider the DiT memory and compute model size by taking the

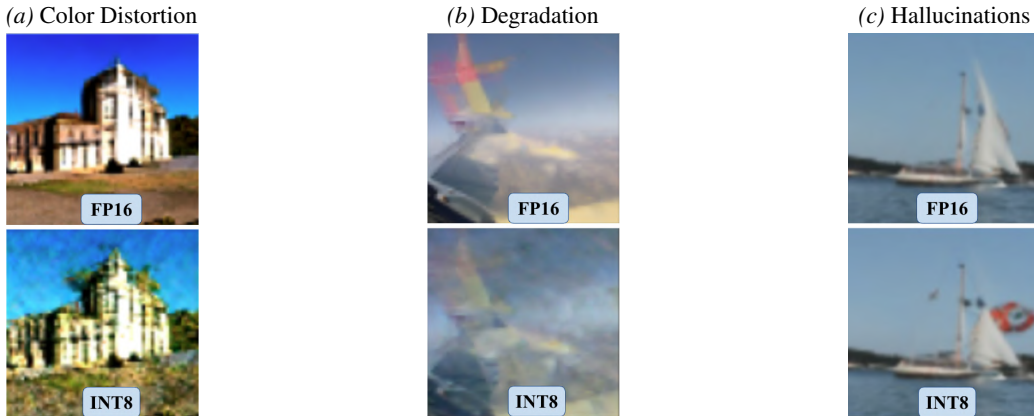


Figure 10. Three types of image artifacts that occur when quantizing image generation models. Images are unconditionally generated from a Two-Step Consistency Model (Song et al., 2023).

Table 9. FP16 Diffusion Model Size Breakdown (in GB)

	LCM	PCM	SDXL-Turbo	FLUX.1[schnell]
UNet/DiT	1.72	4.84	1.03	4.76
Text Encoder(s)	0.25	0.29	0.33	1.95
VAE Decoder	0.07	0.13	0.02	0.02
Total	2.04 GB	5.26 GB	1.37 GB	6.73 GB

parameter count and multiplying it by the number of bytes required per parameter. For W4A4 + LoRA 64, the setup used for SVDQuant (Li et al., 2025), we compute the number of LoRA parameters using the back-of-the-envelope calculation provided in SVDQuant and add it to this calculation. We provide raw data for clarity in Table 10.

Table 10. DiT Memory (in GB) for various bitwidths.

Precision	SDXL-Turbo	FLUX.1[schnell]
FP16	1.03	4.76
W8A8	0.51	2.38
W4A4 + LoRA 64	0.28	1.24
W4A8	0.26	1.19

In Table 11, we provide latency on an RTX A6000 on all models following the same benchmarking procedure as in Table 1.

Table 11. Latency on Nvidia RTX A6000 in milliseconds. All models are evaluated in their 4-step setting.

	FLUX.1	SDXL	SDv1.5	SDXL-Turbo
bfloat16	3.881	0.640	0.193	0.252
w4a8	3.372	0.625	0.172	0.190
w4a8 Q-Sched	3.256	0.621	0.172	0.191

Prior work has already benchmarked latency for SVDQuant on FLUX.1[schnell] and MixDQ on SDXL-Turbo. Here, we directly summarize those experiments. In Tables 12 and 13, INT4 and INT8 are the models that we apply Q-Sched with no additional overhead.

Table 12. Reported in SVDQuant (Li et al., 2025) and summarized here for easy reference. All measurements are for FLUX.1[schnell] on an RTX 4090.

Method	Latency (ms)
BF16	657
INT8	282
INT4	212
SVDQuant	250
SVDQuant + Nunchaku	218

Table 13. Latency for SDXL-Turbo on RTX 4080. These results are reported in MixDQ (Zhao et al., 2024a) and summarized here for easy reference.

Method	Latency (ms)
BF16	24
INT8	16

H. Additional Analysis on COCO-30k

This result reinforces the core finding of our paper: quantization, when paired with a scheduler designed to account for noise sensitivity (as in Q-Sched), can be synergistic with few-step diffusion rather than detrimental. Notably, our quantized model achieves a lower FID than the original full-precision model, suggesting that Q-Sched helps overcome limitations introduced by both step reduction and bit-level compression.

These findings complement the results on SDXL-Turbo and FLUX.1[schnell] discussed in the main paper, and further establish Q-Sched as a general-purpose solution for high-fidelity, compressed diffusion generation.

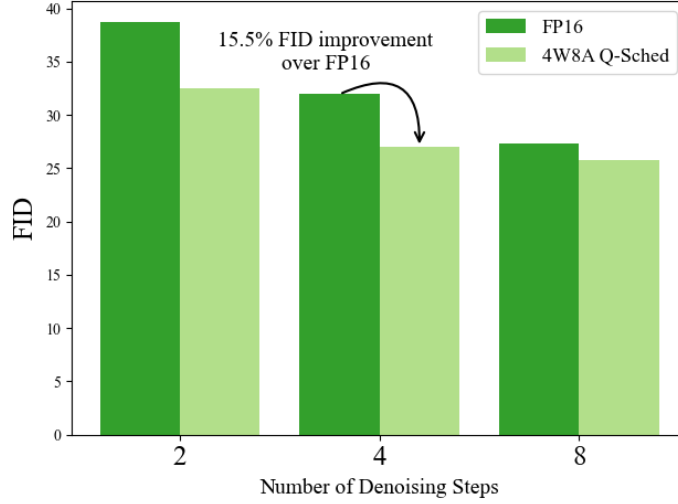


Figure 11. FID on COCO-30k. A W4A8 compressed model with our Q-Sched scheduler outperforms its FP16 counterpart with a 4× reduction in model size.

I. Applying PTQD to the TCD Scheduler

Using PTQD’s linear parameterization for the quantization error, we substitute $\mathcal{E}_\theta^Q(x_t, t) = (1 + \gamma) \cdot \mathcal{E}_\theta + \delta$ into Equation (2):

$$\mathbf{x}_s = \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{\mathbf{x}_t - \sigma_t \mathcal{E}_\theta(x_t, t)}{\alpha_t} + \sigma_{s'} \mathcal{E}_\theta(x_t, t) \right) + \frac{\alpha_s}{\alpha_{s'}(1 + \gamma)} \left(\sigma_{s'} - \frac{\alpha_{s'} \sigma_t}{\alpha_t} \right) \delta + \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \mathbf{z}. \quad (6)$$

PTQD assumes the uncorrelated noise is sampled from a normal distribution $\delta \sim N(\mu_\delta, \sigma_\delta)$. This method applies bias correction to handle the mean deviation, μ_δ , and analytically compute standard deviation, σ_δ . We adapt PTQD’s approach to the TCD schedule and use the new standard deviation, σ_δ for sampling δ :

$$\sigma_\delta^2 = 1 - \frac{\alpha_s^2}{\alpha_{s'}^2} \left(1 - \left(\frac{\delta(\sigma_{s'} - \frac{\alpha_{s'} \sigma_t}{\alpha_t})}{(1 + \gamma)} \right)^2 \right). \quad (7)$$

For the edge case, where $\sigma_\delta < 0$, the deviation is set to zero ($\sigma_\delta = 0$). The proof for extending PTQD to the TCD scheduler is in the appendix.

PTQD attempts to model the distribution shift from a full precision to quantized model using two assumptions:

1. The quantized model’s distribution shift can be modeled through a linear correction term.
2. The uncorrelated quantization noise is normally distributed.

While these assumptions are similar to prior work on diffusion models, they are likely to break down on the few-step

diffusions where the denoising process is distilled from many steps and is not expected to be linear nor follow a Gaussian distribution.

Quantization Noise Correction using PTQD Based on the PTQD quantization noise assumption, the quantization error is linearly parametrized as $\Delta\mathcal{E}_\theta = \gamma \cdot \mathcal{E}_\theta + \delta$ where γ, δ are learnable parameters corresponding to the correlated noise w.r.t. full precision and the uncorrelated noise respectively. PTQD models the uncorrelated noise as Gaussian (*i.e.*, $\delta \sim \mathcal{N}(\mu_q, \sigma_q)$).

Variance Schedule Calibration for Trajectory Consistency Distillation (TCD) TCD’s Strategic Stochastic Sampling (SSS) using a quantized network $\mathcal{E}_\theta^Q(x_t, t)$ is given by:

$$\mathbf{x}_s = \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{\mathbf{x}_t - \sigma_t \mathcal{E}_\theta^Q(x_t, t)}{\alpha_t} + \sigma_{s'} \mathcal{E}_\theta^Q(x_t, t) \right) + \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \mathbf{z} \quad (8)$$

Using PTQD’s linear parametrization for the quantization error, we substitute $\mathcal{E}_\theta^Q(x_t, t) = (1 + \gamma) \cdot \mathcal{E}_\theta + \delta$:

$$\mathbf{x}_s = \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{\mathbf{x}_t - \sigma_t ((1 + \gamma) \cdot \mathcal{E}_\theta(x_t, t) + \delta)}{\alpha_t} + \sigma_{s'} ((1 + \gamma) \cdot \mathcal{E}_\theta(x_t, t) + \delta) \right) + \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \mathbf{z} \quad (9)$$

$$= \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{\mathbf{x}_t - \sigma_t (1 + \gamma) \mathcal{E}_\theta(x_t, t)}{\alpha_t} + \sigma_{s'} (1 + \gamma) \mathcal{E}_\theta(x_t, t) - \frac{\alpha_{s'} \sigma_t \delta}{\alpha_t} + \sigma_{s'} \delta \right) + \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \mathbf{z} \quad (10)$$

$$= \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{\mathbf{x}_t - \sigma_t (1 + \gamma) \mathcal{E}_\theta(x_t, t)}{\alpha_t} + \sigma_{s'} (1 + \gamma) \mathcal{E}_\theta(x_t, t) \right) + \frac{\alpha_s}{\alpha_{s'}} \left(\sigma_{s'} - \frac{\alpha_{s'} \sigma_t}{\alpha_t} \right) \delta + \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \mathbf{z} \quad (11)$$

The correlated noise can be corrected by applying:

$$\frac{\mathcal{E}_\theta^Q(x_t, t)}{1 + \gamma} = \frac{(1 + \gamma) \mathcal{E}_\theta(x_t, t) + \delta}{1 + \gamma} \quad (12)$$

$$= \mathcal{E}_\theta(x_t, t) + \frac{\delta}{1 + \gamma} \quad (13)$$

The resultant SSS sampling step becomes:

$$\mathbf{x}_s = \frac{\alpha_s}{\alpha_{s'}} \left(\alpha_{s'} \frac{\mathbf{x}_t - \sigma_t \mathcal{E}_\theta(x_t, t)}{\alpha_t} + \sigma_{s'} \mathcal{E}_\theta(x_t, t) \right) + \frac{\alpha_s}{\alpha_{s'} (1 + \gamma)} \left(\sigma_{s'} - \frac{\alpha_{s'} \sigma_t}{\alpha_t} \right) \delta + \sqrt{1 - \frac{\alpha_s^2}{\alpha_{s'}^2}} \mathbf{z} \quad (14)$$

The variance schedule becomes:

$$\sigma_\delta^2 = 1 - \frac{\alpha_s^2}{\alpha_{s'}^2} - \left(\frac{\alpha_s}{\alpha_{s'} (1 + \gamma)} \left(\sigma_{s'} - \frac{\alpha_{s'} \sigma_t}{\alpha_t} \right) \right)^2 \delta^2 \quad (15)$$

$$= 1 - \frac{\alpha_s^2}{\alpha_{s'}^2} \left(1 - \frac{(\sigma_{s'} - \frac{\alpha_{s'} \sigma_t}{\alpha_t})^2}{(1 + \gamma)^2} \delta^2 \right) \quad (16)$$

$$= 1 - \frac{\alpha_s^2}{\alpha_{s'}^2} \left(1 - \left(\frac{\delta (\sigma_{s'} - \frac{\alpha_{s'} \sigma_t}{\alpha_t})}{(1 + \gamma)} \right)^2 \right) \quad (17)$$

Since $\mathbf{z} \sim N(\mu_\delta, \sigma_\delta)$, we must handle the edge case when $\sigma_\delta < 0$. If the variance is negative, we simply set $\sigma_\delta = 0$.

Upon comparing Q-Sched to PTQD you may ask ”Why is Q-Sched able to learn a better noise schedule when it is also a linear correction?” Q-Sched learns scalar coefficients on x_t and $\mathcal{E}_\theta$ that are optimized with respect to the reference-free

JAQ loss. This allows us to learn a new schedule with linear corrections to improve our overall noise schedule, rather than matching the existing full precision schedule. This is an important distinction from PTQD, which tries to learn a linear correction with respect to full precision, which may not be possible since quantization produces a nonlinear distortion on the diffusion model. In short, PTQD attempts to match the full precision sampling trajectory, whereas Q-Sched aims to learn a new sampling trajectory given a compressed $\mathcal{E}_\theta$.

J. Proof of Theorem 1: Strict Existence Guarantees for Quantization-Aware Scheduling

Theorem 3.1 (Strict Existence Guarantees). There exist Q-Sched coefficients $(c^\epsilon, c^x) \neq 0$ such that $E[||\Delta\tilde{x}_0||] < E[||\Delta x_0||]$.

Proof. Let us consider the few-step sampling trajectories for the pre-trained and quantized models, parametrized by $\mathcal{E}_\theta(t)$ and $\mathcal{E}_\theta^Q(t)$ respectively. These two few-step diffusion models sample at the same time-steps, $0 = t_0 < t_1, t_2 \dots t_N = T$, where N represents the number of steps in the few-step model. For ease of notation, we will use the time-step 0 to refer to t_0 and 1 to refer to t_1 , etc. A denoising step going from time $t+1 \rightarrow t$, produces a partially denoised image, x_t , and its quantized counterpart, x_t^Q . Following directly from Equation 9, the denoising error, $\Delta x_t = x_t - x_t^Q$, can be explicitly computed as:

$$\Delta x_t = \frac{\alpha_t}{\alpha_{t'}} \left(\alpha_{t'} \frac{\Delta x_{t+1} - \sigma_{t+1}(\mathcal{E}_\theta(t+1) - \mathcal{E}_\theta^Q(t))}{\alpha_{t+1}} + \sigma_{t'}(\mathcal{E}_\theta(t+1) - \mathcal{E}_\theta^Q(t+1)) \right) \quad (18)$$

$$= \frac{\alpha_t}{\alpha_{t+1}} \Delta_{t+1} + \frac{\alpha_t}{\alpha_{t'}} \left(\sigma_{t'} - \frac{\sigma_{t+1}}{\alpha_{t+1}} \right) (\mathcal{E}_\theta(t+1) - \mathcal{E}_\theta^Q(t+1)) \quad (19)$$

$$= k_t \Delta x_{t+1} + m_t \Delta \mathcal{E}_\theta(t+1) \quad (20)$$

where we define the sampler coefficients as $k_t = \frac{\alpha_t}{\alpha_{t+1}}$, $m_t = \frac{\alpha_t}{\alpha_{t'}} \left(\sigma_{t'} - \frac{\sigma_{t+1}}{\alpha_{t+1}} \right)$ and denote the change in the network as $\Delta \mathcal{E}_\theta(t) = \mathcal{E}_\theta(t) - \mathcal{E}_\theta^Q(t)$. Assuming the initial denoised image is the same ($x_N = x_N^Q$), the error in the final denoised image, Δx_0 , is given by:

$$\Delta x_0 = k_0 \Delta x_1 + m_0 \Delta \mathcal{E}_\theta(1) \quad (21)$$

$$= k_0 k_1 k_2 \dots (k_N \Delta x_N + m_{N-1} \Delta \mathcal{E}_\theta(N)) + \dots + k_0 k_1 m_2 \Delta \mathcal{E}_\theta(3) + k_0 m_1 \Delta \mathcal{E}_\theta(2) + m_0 \Delta \mathcal{E}_\theta(1) \quad (22)$$

$$= \sum_{s=1}^S \left(\prod_{v=0}^{s-2} k_v \right) m_{s-1} \Delta \mathcal{E}_\theta(s) \quad (23)$$

The average expected error over all images in a given dataset, $x_0 \in \mathcal{D}$, is given by:

$$E[||\Delta x_0||] = \sum_{s=1}^S \left(\prod_{v=0}^{s-2} k_v \right) m_{s-1} E[||\Delta \mathcal{E}_\theta(s)||] \quad (24)$$

since $E[||\Delta x_0||]$ is a homogeneous function.

In Q-Sched, we apply our quantization-aware pre-conditioning on every noise coefficient: $\tilde{m}_t = c_t^\epsilon \cdot m_t$ and $\tilde{k}_t = c_t^x \cdot k_t$. Let us denote the expected error induced by Q-Sched with respect to the pre-trained model's x_0 as $E[||\Delta \tilde{x}_0||]$.

We empirically show in Tables 1 and 2 that $E[||\Delta x_0||] \neq 0$ since the images produced by the naive quantization method produce a different FID from the original pre-trained model's image distribution. Since Equation 24 is a linear function of $k_t, m_t, \forall t \in 1 \dots N$, and there is a global minimum at $E[||x_0 - x_0||] = 0$, it must be that $\exists \tilde{m}_t^*, \tilde{k}_t^* \forall t$ such that $E[||\Delta \tilde{x}_0||] < E[||\Delta x_0||]$. In short, we guarantee that there exists quantization-aware coefficients that strictly improve our expected quantization error over naive quantization. $\square$

Aside: Positive Sampler Coefficients The TCD Scheduler has $\beta_0 = 0.0085, \beta_N = 0.012, \alpha_t = 1 - \beta_t, \sigma_t = \prod_{i=0}^t \alpha_i$ with a scaled linear schedule:

$$\beta_t = \left(\sqrt{\beta_0} + t \cdot (\sqrt{\beta_N} - \sqrt{\beta_0}) \right)^2 \quad (25)$$

Therefore, $1 > \alpha_0 > \alpha_1 > \dots > \alpha_N > 0$ and $1 > \sigma_0 > \sigma_1 > \dots > \sigma_N > 0$. We note the $t' = (1 - \gamma)t$ where $\gamma \in [0, 1]$, so $t' \leq t$. This implies that $\sigma_{t'} > \sigma_{t+1}$ so:

$$k_t > 0, \quad m_t = \frac{\alpha_t}{\alpha_{t'}} \left(\sigma_{t'} - \frac{\sigma_{t+1}}{\alpha_{t+1}} \right) > 0 \quad (26)$$

This illustrates that $k_t, m_t \in \mathbb{R}^+$.

K. Scheduler coefficients affect final image quality

Two coefficients are expressive enough for modest compression scenarios where quantization causes a mild drift in the sampling trajectory. Adjusting the relative contribution of the predicted noise versus the intermediate state is typically enough to realign this trajectory. We observe catastrophic error when quantizing further to 3-bit integer weights and in contrast, observe only modest gains in the 8-bit scenario across both SDXL-Turbo and FLUX.1 [schnell]. We observe that Q-Sched works best in the 4-bit setting, where compression yields noticeable degradation but is not as effective in the 3-bit setting, where the generated images are very poor quality.

In Figure 12, we illustrate how three different coefficient combinations can yield images with unique artifacts and details. We take five images generated from FLUX.1[schnell] with different sampler coefficients parameterized by (c_{min}, c_{max}) .

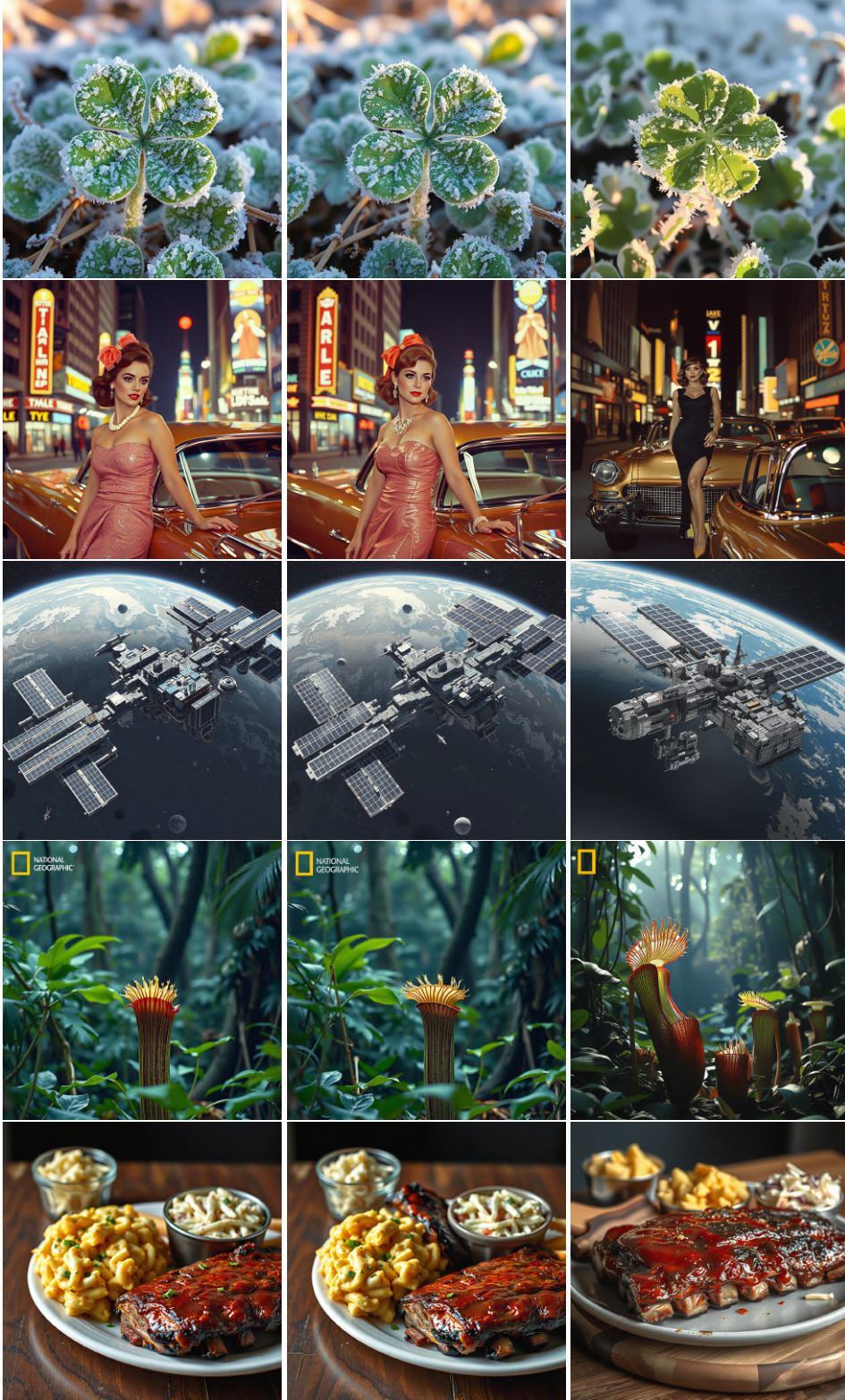


Figure 12. Qualitative comparison of generated samples across three coefficient settings. Each row corresponds to a consistent prompt, random seed and model. From left to right, coefficients (c_{min}, c_{max}) are $(0.8, 0.933)$, $(0.843, 0.909)$, and $(1.10, 0.82)$.

L. Ablation on TC () vs. IQ ()

In Table 14, we show that optimizing Q-Sched with respect to TC () or IQ () alone improves some metrics but not others. For example, optimizing purely for text compatibility with AQ-MAP yields strong CLIP Score and Pick Score, but noticeably lower CLIP-IQA and HPV2. In contrast, optimizing with JAQ produces consistent gains across both image quality and text compatibility. We see improvements in CLIP Score and CLIP-IQA while still maintaining strong performance on the text compatibility metrics.

Table 14. Comparison of scheduler objectives across metrics.

Metric	AQ-MAP	HPV2	JAQ
CLIP Score $\uparrow$	25.266	25.043	25.269
CLIP-IQA $\uparrow$	0.698	0.705	0.707
HPV2 $\uparrow$	0.271	0.278	0.272
AQ-MAP $\uparrow$	0.701	0.696	0.700
JAQ (ours) $\uparrow$	1.673	1.669	1.673
Pick Score $\uparrow$	18.50	18.38	18.46

M. Scalability to Other Types of Diffusions

In Table 1, we display scalability across 2,4, and 8-step consistency models and have every indication that our method will scale to longer trajectories. Yet, we emphasize that the scope of our work is to consider few-step diffusion models, which are typically less than 8 steps.

Following the same setting as in Table 1, we also provide a single run on PCM with 16 steps, illustrating exciting performance and the potential for Q-Sched to improve on longer step regimes:

Table 15. Q-Sched on 16-Step Phased Consistency Model, illustrating scalability to a larger number of steps.

16-step	FID
FP16	19.829
INT4	19.567
INT4 Q-Sched	15.969

Why not other types of few-step models? One popular few-step diffusion model, SANA-sprint (Chen et al., 2025), is a much smaller model (0.6B parameters) than SDXL-Turbo or FLUX.1, so quantization offers limited benefit and tends to cause more degradation at this scale. Q-Sched is most impactful for distilled, few-step models where size meaningfully affects inference speed, making SANA-sprint a less natural target. While Q-Sched could be applied, we leave this direction for future work.

N. Optimization Details

Optimizing scheduler coefficients independently at each timestep is a natural baseline, but it induces a prohibitively large search space that grows exponentially with the number of denoising steps. Instead, Q-Sched parameterizes the scheduler using only two coefficients per variable, (c_{start}, c_{end}) for both c^x and c^ϵ , and applies linear spacing across timesteps. This parameterization avoids an exponentially large search space while yielding strong empirical performance.

Under a naive formulation, the search complexity scales as $O(n^\omega \cdot |\mathcal{C}|)$, where:

- n is the number of candidate values per coefficient,
- ω is the number of diffusion steps, and
- $|\mathcal{C}|$ is the calibration dataset size.

As the number of steps increases, grid search over per-timestep coefficients quickly becomes intractable. As shown in Algorithm 1, Q-Sched removes the exponential dependence on ω by learning only the start and end coefficients and deterministically interpolating intermediate values. This reduces the effective search space to a constant dimensionality, benefiting both grid search and evolutionary optimization methods. As a result, Q-Sched scales linearly with the calibration set size and remains independent of the number of denoising steps.